

# 2.3 Group 17

## Question Paper

|            |                        |
|------------|------------------------|
| Course     | CIEA Level Chemistry   |
| Section    | 2. Inorganic Chemistry |
| Topic      | 2.3 Group 17           |
| Difficulty | Easy                   |

**Time allowed:** 20

**Score:** /10

**Percentage:** /100

### Question 1

What happens when iodine is bubbled through aqueous potassium bromide?

- A. Iodine is oxidised to iodide ions.
- B. Potassium bromide is reduced to bromine.
- C. Bromide ions are oxidised to bromine.
- D. No reaction occurs.

[1 mark]

### Question 2

More than half a million tonnes of bromine is produced annually and is largely used for making other compounds. One important use of bromine is to manufacture agricultural chemicals.

What is another important use for bromine?

- A. Flame-retardants and fire extinguishers
- B. Water purification
- C. Antiseptic agents
- D. Bleaches for textiles and the paper industry

[1 mark]

### Question 3

Chlorine is widely used in water treatment plants. Chlorine is reacted with water to produce hydrochloric acid and another chlorine-containing acid.

Which row correctly describes the properties of the reaction?

|   | Moles $\text{Cl}_2$ used | Moles $\text{HCl}$ produced | Other product formed |
|---|--------------------------|-----------------------------|----------------------|
| A | 1                        | 1                           | $\text{HClO}$        |
| B | 2                        | 3                           | $\text{HClO}_2$      |
| C | 3                        | 5                           | $\text{HClO}_3$      |
| D | 4                        | 7                           | $\text{HClO}_4$      |

[1 mark]

### Question 4

The halogens exist as diatomic molecules,  $X_2$ .

Descending through Group 17 from chlorine to iodine the boiling points of the elements increase.

Which statement explains this trend?

- A. Of the permanent dipole in the  $X_2$  molecule increases as the group is descended.
- B. The X-X bond strength increases as the group is descended.
- C. the electronegativity of X decreases as the group is descended.
- D. The number of electrons in each  $X_2$  molecule increases as the group is descended.

[1 mark]

### Question 5

How do the strengths of the forces between molecules, and the bonds within molecules, vary going down Group 17 from chlorine to bromine to iodine?

|   | Strength of van der Waals' forces | Strength of covalent bonds |
|---|-----------------------------------|----------------------------|
| A | Increase                          | Increase                   |
| B | Decrease                          | Increase                   |
| C | Increase                          | Decrease                   |
| D | Decrease                          | Decrease                   |

[1 mark]

### Question 6

What happens when chlorine gas is bubbled through aqueous potassium iodide?

- A. Iodide ions are oxidised to iodine.
- B. Chlorine is oxidised to chlorate (V) ions.
- C. Chlorine is oxidised to chloride ions.
- D. There is no observable reaction.

[1 mark]

### Question 7

A scientist compares the properties of chlorine, iodine and their compounds.

Property **X** for iodine is bigger than for chlorine.

What is property **X**?

- A. Solubility of the silver halide in  $\text{NH}_3$  (aq)
- B. Oxidising ability of the element
- C. Thermal stability of the hydrogen halide
- D. Strength of van der Waals' forces between the molecules of the element

[1 mark]

### Question 8

Which properties of hydrogen halides steadily increase in the sequence  $\text{HCl}$ ,  $\text{HBr}$  and  $\text{HI}$ ?

- 1 Bond length
- 2 Ease of oxidation
- 3 Thermal stability

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

**Question 9**

When halogen sodium salts react with concentrated sulfuric acid which of the halide ions, chloride, bromide or iodide, acts as a reducing agent?

1  $\text{Cl}^-$ 2  $\text{Br}^-$ 3  $\text{I}^-$ 

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

**[1 mark]****Question 10**

When a hot platinum wire is plunged into a test tube of hydrogen chloride, no change occurs. However, if the experiment is repeated with hydrogen iodide, the gas is decomposed into its elements.

Which factors result in this behaviour?

1 The strength of the hydrogen-halogen bond.

2 The standard enthalpy of formation,  $\Delta H_f^\circ$ , of each of the products of decomposition.

3 The size of the halogen atom.

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

**[1 mark]**

## Model Answers: Easy

1

The correct answer is **D** because:

- The iodine will not displace the bromine.
- Bromine is higher up in Group 17, therefore, more reactive and only halogens that are higher up will displace halogens that are lower in the group.

2

The correct answer is **A** because:

- Bromine compounds are mainly used in flame retardant materials and fire extinguishers.

**B** is incorrect as chlorine is the main halogen used for water purification.

**C** is incorrect as iodine is the main halogen used as an antiseptic agent.

**D** is incorrect as chlorine is the main halogen used as a bleaching agent.

3

The correct answer is **A** because:

- The reaction of water with chlorine proceeds very slowly.
  - $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HClO}$
- This shows that one mole of  $\text{Cl}_2$  makes one mole of  $\text{HCl}$  and one mole of  $\text{HClO}$ .

4

The correct answer is **D** because:

- There are only weak van der Waals' forces between the diatomic molecules,

caused by instantaneous dipole-induced dipole forces.

- These forces increase as you go down Group 17 as the number of electrons in the molecules increase.
- The greater the number of electrons the greater the chance of instantaneous dipoles arising within molecules inducing dipoles in neighbouring molecules.
- The larger the molecules, the stronger the van der Waals' forces.
- Therefore, iodine has a stronger force than fluorine.
- The stronger the force, the more energy needed to break the molecules from each other and therefore the boiling points would increase.

**A** is incorrect as the dipoles are not permanent.

**B** is incorrect as the bond energy decreases as you go down Group 17.

**C** is incorrect as the electronegativity affects the reactivity of the Group 17 elements.

5

The correct answer is **C** because:

- There are only weak van der Waals' forces between the diatomic molecules, caused by instantaneous dipole-induced dipole forces.
- These forces increase as you go down Group 17 as the number of electrons in the molecules increase.
- The greater the number of electrons the greater the chance of instantaneous dipoles arising within molecules inducing dipoles in neighbouring molecules.
- The larger the molecules, the stronger the van der Waals' forces.
- Therefore, the forces increase as you go down the group.
- The strength of a covalent bond is determined by the strength of the attractions between the bonding pair of electrons and the two nuclei.
- Fluorine is a smaller molecule with less shielding so has the strongest pull on the pair of electrons in a covalent bond.

- The bond energy decreases as you go down the group.

6

The correct answer is **A** because:

- The chlorine will displace the iodine.
- Chlorine is higher up in Group 17, therefore, more reactive and will displace halogens that are lower in the group.
- Remember OILRIG; oxidation is the loss of electrons, and reduction is gain of electrons.
- Iodine ions will have an extra electron to be  $I^-$  so will be oxidised to iodine.

**B** is incorrect as the chlorate (V) ion structure is  $ClO_3^-$ , there are no oxygen atoms in this reaction to make the chlorate ion.

**C** is incorrect as chlorine is more reactive than iodine so displacing the iodine from the potassium.

**D** is incorrect as chlorine is more reactive there will be a displacement reaction.

7

The correct answer is **D** because:

- There are only weak van der Waals' forces between the diatomic molecules, caused by instantaneous dipole-induced dipole forces.

pair of electrons in a covalent bond.

- The bond energy decreases as you go down the group.

6

The correct answer is **A** because:

- The chlorine will displace the iodine.
- Chlorine is higher up in Group 17, therefore, more reactive and will displace halogens that are lower in the group.
- Remember OILRIG; oxidation is the loss of electrons, and reduction is gain of electrons.
- Iodine ions will have an extra electron to be  $I^-$  so will be oxidised to iodine.

**B** is incorrect as the chlorate (V) ion structure is  $ClO_3^-$ , there are no oxygen atoms in this reaction to make the chlorate ion.

**C** is incorrect as chlorine is more reactive than iodine so displacing the iodine from the potassium.

**D** is incorrect as chlorine is more reactive there will be a displacement reaction.

7

The correct answer is **D** because:

- There are only weak van der Waals' forces between the diatomic molecules, caused by instantaneous dipole-induced dipole forces.
- These forces increase as you go down Group 17 as the number of electrons in the molecules increase.
- The greater the number of electrons the greater the chance of instantaneous dipoles arising within molecules inducing dipoles in neighbouring molecules.
- The larger the molecules, the stronger the van der Waals' forces.
- Therefore, the forces increase as you go down the group.

**A** is incorrect as the solubility of the silver halides in  $\text{NH}_3$  decreases going down the group silver iodide is not soluble in ammonia whereas silver chloride is.

**B** is incorrect as the halogen atoms get larger, they accept electrons less easily, and the oxidising power becomes weaker.

**C** is incorrect as the thermal stability of the hydrogen halides decreases as you go down the group. This is because the hydrogen halide bond is weaker.

8

The correct answer is **B** because:

- The bond length increases as you go down Group 17, this is because the bond length is proportional to the ionic radius.
- The ionic radius increases as you down Group 17.
- The larger the halide ion, the further the outer electrons are from the nucleus and the more they are shielded by inner electrons.
- It, therefore, gets easier for the halide ions to lose electrons down the group and become oxidised.
  - Remember OILRIG; oxidation is the loss of electrons, and reduction is gain of electrons.

Statement 3 is incorrect as the thermal stability of the halogen halides decreases as you go down Group 17.

9

The correct answer is **C** because:

- In this reaction, the concentrated sulfuric acid can act both as an acid and an oxidising agent.
- When it acts as an acid it produces hydrogen halide and sodium hydrogen sulfate (if the reacting halogen was a halogen sodium salt).
  - All the halogens react in this way.
- When it acts as an oxidising agent, meaning the halogen is acting as a reducing

agent.

- Only bromine and iodine are strong enough reducing agents to reduce the concentrated sulfuric acid.
  - For example, in sodium bromide; the bromide ions are oxidised to bromine.
  - The bromide ions reduce the sulfuric acid to sulfur dioxide gas.

Fluoride and chloride ions won't reduce concentrated sulfuric acid as they are not strong enough reducing agents.

10

The correct answer is **D** because:

- The **thermal stability** of the hydrogen halides decreases as you go down the group. This is because the hydrogen halide bond is weaker.
- The bond gets weaker as you go down the group as the **halogen atoms are getting bigger** and the bond length is longer, and the bonding pair of electrons is getting further from the halogen nucleus.
- **The standard enthalpy of formation:**
  - The enthalpy change when 1 mole of a compound is formed from its elements with both the elements and the compound being in their standard states.
- The enthalpy of formation of iodine is less than chlorine and therefore will need less energy once decomposed to the ions to form the diatomic elements.

